Number Systems

Decimal Numbers

- This is the most intuitive numbering systems for humans.
- It uses a base of ten
- $157_{10} = 1\times10^2 + 5\times10^1 + 7\times10^0$
- $1760_{10} = 1x10^3 + 7\times10^2 + 6\times10^1 + 0\times10^0$
- Advantages:
- Intuitive and easy to work with for humans
- We can represent relatively large numbers with a few digits
- Disadvantage (from a computing perspective)
- It is not very easy to convert to binary numbers (because not all the multiples of 2 are also multiples of 10).

Binary Numbers

- **Binary Number System:** a binary number is defined as a number that is expressed in the binary system or base 2 numeral system.
- It describes numeric values by two separate symbols; 1 (one) and 0 (zero). The base-2 system is the positional notation with 2 as a radix.
- The binary system is applied internally by almost all latest computers and computer-based devices because of its direct implementation in electronic circuits using logic gates.
- Every digit is referred to as a **bit**.
- $101_2 = 1\times2^2 + 0\times2^1 + 1\times2^0$
- Addition and subtraction

$$\begin{array}{r} 1011 \\ +\ 1010 \\ \hline 10101 \end{array} \qquad \begin{array}{r} 1011 \\ -\ 0110 \\ \hline 0101 \end{array}$$

Octal Numbers

- Octal Number System is one the type of Number Representation techniques, in which there value of base is 8.
- That means there are only 8 symbols or possible digit values, there are 0, 1, 2, 3, 4, 5, 6, 7.
- It requires only 3 bits to represent value of any digit.
- Octal numbers are indicated by the addition of either an 0o prefix or an 8 suffix.
- $63_8 = 6\times8^1 + 3\times8^0$
- **Advantage**: The main advantage of using Octal numbers is that it uses less digits than decimal and Hexadecimal number system.
- So, it has fewer computations and less computational errors. It uses only 3 bits to represent any digit in binary and easy to convert from octal to binary and vice-versa.
- It is easier to handle input and output in the octal form.
- **Disadvantage**: of Octal number system is that computer does not understand octal number system directly, so we need octal to binary converter.

Hexadecimal Numbers

- Hexadecimal Number System is one the type of Number Representation techniques, in which there value of base is 16.
- That means there are only 16 symbols or possible digit values, there are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F.
- Where A, B, C, D, E and F are single bit representations of decimal value 10, 11, 12, 13, 14 and 15 respectively.
- It requires only 4 bits to represent value of any digit.
- Hexadecimal numbers are indicated by the addition of either an *0x* prefix or an *h* suffix.
- $A1_{16} = 10 \times 16^1 + 1 \times 16^0$
- **Advantage:** The main advantage of using Hexadecimal numbers is that it uses less memory to store more numbers, for example it store 256 numbers in two digits whereas decimal number stores 100 numbers in two digits.
- **Disadvantage**: is that it may not an easy to read and write for people, and also difficult to perform operations like multiplications, divisions using hexadecimal number system.

Binary → hex/decimal/octal conversion

- Conversion from binary to octal/hex
  - Binary: 10011110001
  - Octal: 10 | 011 | 110 | 001=2361_8
  - Hex: 100 | 1111 | 0001=$4F1_{16}$
- Conversion from binary to decimal
  - $101_2 = 1\times2^2 + 0\times2^1 + 1\times2^0 = 5_{10}$
  - $63.4_8 = 6\times8^1 + 3\times8^0 + 4\times8^{-1} = 51.5_{10}$
  - $A1_{16} = 10\times16^1 + 1\times16^0 = 161_{10}$

Decimal⟶ binary/octal/hex conversion

Binary

	Quotient	Remainder
56÷2=	28	0
28÷2=	14	0
14÷2=	7	0
7÷2=	3	1
3÷2=	1	1
1÷2=	0	1

Octal

	Quotient	Remainder
56÷8=	7	0
7÷8=	0	7

$56_{10}=111000_2$

$56_{10}=70_8$

- Why does this work?
  - $N=56_{10}=111000_2$
  - Q=N/2=56/2=111000/2=11100 remainder 0
- Each successive divide liberates an LSB (least significant bit)

Negative Numbers

- How do we write negative binary numbers?
- Historically: 3 approaches
  - Sign-and-magnitude
  - Ones-complement
  - Twos-complement
- For all 3, the most-significant bit (MSB) is the sign digit
  - 0 ≡ positive
  - 1 ≡ negative
- twos-complement is the important one
  - Simplifies arithmetic
  - Used almost universally

Sign-and-magnitude

- The most-significant bit (MSB) is the sign digit
  - 0 ≡ positive
  - 1 ≡ negative
- The remaining bits are the number's magnitude
- Problem 1: Two representations for zero
  - 0 = 0000 and also –0 = 1000
- Problem 2: Arithmetic is cumbersome

Add		Subtract			Compare and subtract		
4	0100	4	0100	0100	– 4	1100	1100
+ 3	+ 0011	– 3	+ 1011	– 0011	+ 3	+ 0011	– 0011
= 7	= 0111	= 1	≠ 1111	= 0001	– 1	≠ 1111	= 1001

Ones-complement

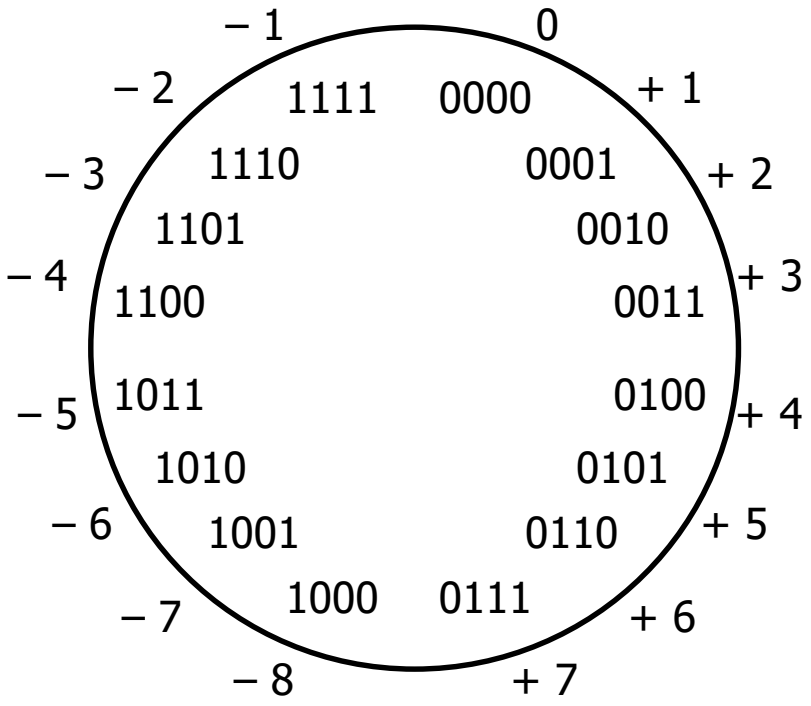
- Negative number: Bitwise complement positive number
  - $0011 \equiv 3_{10}$
  - $1100 \equiv -3_{10}$
- Solves the arithmetic problem

Add		Invert, add, add carry		Invert and add	
4	0100	4	0100	– 4	1011
+ 3	+ 0011	– 3	+ 1100	+ 3	+ 0011
= 7	= 0111	= 1	1 0000	– 1	1110
		add carry:	+1		
			= 0001		

- Remaining problem: Two representations for zero
  - 0 = 0000 and also –0 = 1111

Twos-complement

- Negative number: Bitwise complement plus one
  - $0011 \equiv 3_{10}$
  - $1101 \equiv -3_{10}$
- Number wheel



- Only one zero!
- MSB is the sign digit
  - 0 ≡ positive
  - 1 ≡ negative

Twos-complement (con't)

- Complementing a complement ➲ the original number
- Arithmetic is easy
  - Subtraction = negation and addition
    - Easy to implement in hardware

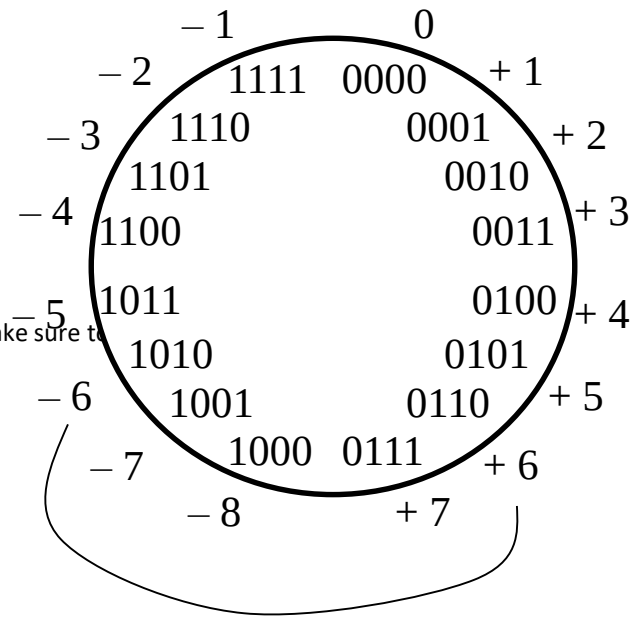
Add		Invert and add		Invert and add	
4	0100	4	0100	– 4	1100
+ 3	+ 0011	– 3	+ 1101	+ 3	+ 0011
= 7	= 0111	= 1	1 0001	– 1	1111
		drop carry	= 0001		

Miscellaneous

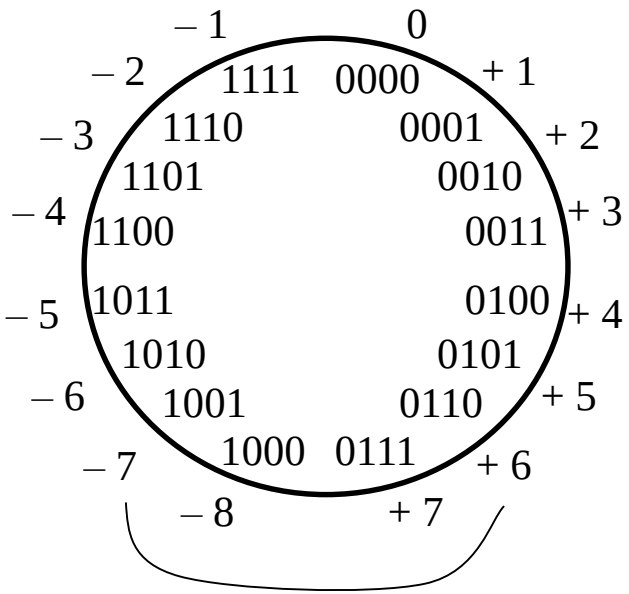
- Twos-complement of non-integers
  - $1.6875_{10} = 01.1011_2$
  - $-1.6875_{10} = 10.0101_2$
- Sign extension
  - Write +6 and –6 as twos complement
    - 0110 and 1010
  - Sign extend to 8-bit bytes
    - 00000110 and 11111010
- Can't infer a representation from a number
  - 11001 is 25 (unsigned)
  - 11001 is –9 (sign magnitude)
  - 11001 is –6 (ones complement)
  - 11001 is –7 (twos complement)

Twos-complement overflow

- Summing two positive numbers gives a negative result
- Summing two negative numbers gives a positive result
- Make sure t



$6 + 4 \Rightarrow -6$



$-7 - 3 \Rightarrow +6$

Gray and BCD codes

Decimal Symbols	Gray Code
0	0000
1	0001
2	0011
3	0010
4	0110
5	0111
6	0101
7	0100
8	1100
9	1101

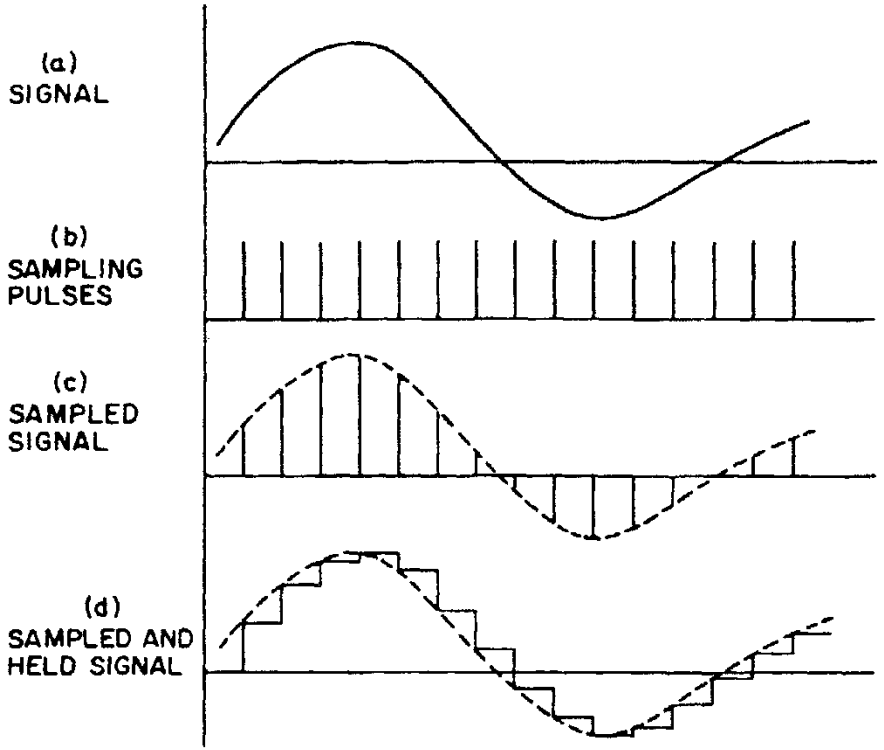
Decimal Symbols	BCD Code
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

The physical world is analog

- Digital systems need to
  - Measure analog quantities
    - Speech waveforms, etc
  - Control analog systems
    - Drive motors, etc
- How do we connect the analog and digital domains?
  - Analog-to-digital converter (A/D)
    - Example: CD recording
  - Digital-to-analog converter (D/A)
    - Example: CD playback

Sampling

- **Quantization**
  - Conversion from analog to discrete values
- Quantizing a signal
  - We sample it

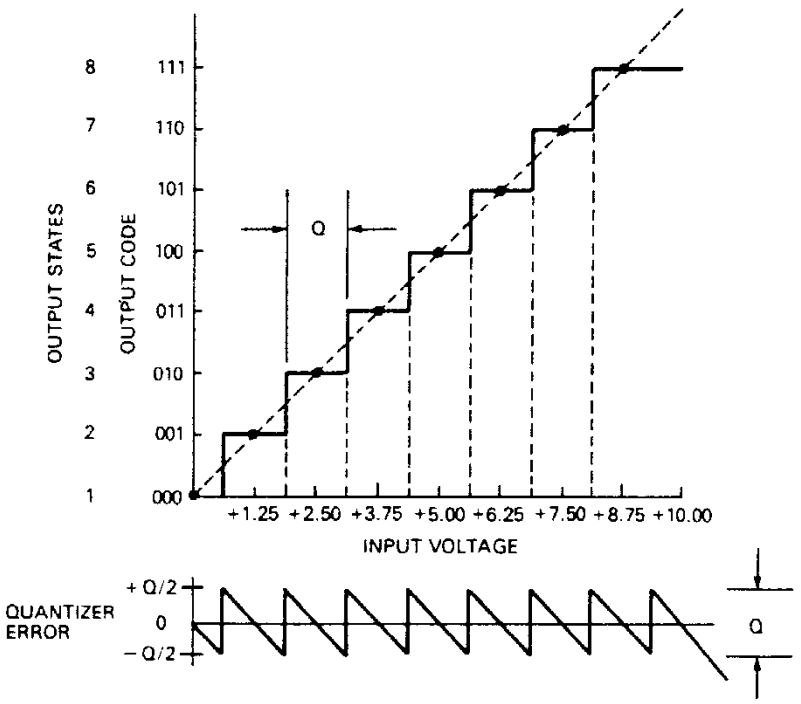


Signal Sampling

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Conversion

- **Encoding**
  - Assigning a digital word to each discrete value
- Encoding a quantized signal
  - Encode the samples
  - Typically Gray or binary codes



Transfer Function of Ideal 3-Bit Quantizer

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